

# CHANGWOON CHOI

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## Education

|                           |                                                 |                   |
|---------------------------|-------------------------------------------------|-------------------|
| Seoul National University | M.S. & Ph.D., Electrical & Computer Engineering | 2020.9 - Present  |
| Seoul National University | B.S., Electrical & Computer Engineering         | 2014.03 - 2020.08 |
| Seoul Science High School |                                                 | 2011.03 - 2014.02 |

## Publications

1. **Changwoon Choi\***, Hyunsoo Lee\*, Clément Jambon, Yael Vinker, and Young Min Kim, *Image-Guided Geometric Stylization of 3D Meshes*, in *CVPR* (2026).
2. Minkwan Kim, **Changwoon Choi**, and Young Min Kim, *Geometry-Aware Scene Configurations for Novel View Synthesis*, in *IEEE TVCG* (2026).
3. Clément Jambon, **Changwoon Choi**, Dongsu Zhang, Olga Sorkine-Hornung, and Young Min Kim, *ExCellGen: Fast, Controllable, Photorealistic 3D Scene Generation from a Single Real-World Exemplar*, in *Computational Visual Media Journal* (2026, Best Paper Honorable Mention).
4. **Changwoon Choi**, Jeongjun Kim, Geonho Cha, Minkwan Kim, Dongyoon Wee, and Young Min Kim, *Humans as a Calibration Pattern: Dynamic 3D Scene Reconstruction from Unsynchronized and Uncalibrated Videos*, in *ICCV* (2025).
5. Jaeah Lee, **Changwoon Choi**, Young Min Kim, and Jaesik Park, *Recovering Dynamic 3D Sketches from Videos*, in *CVPR* (2025).
6. Gwangtak Bae\*, **Changwoon Choi\***, Hyeongjun Heo, Sang Min Kim, and Young Min Kim, *I<sup>2</sup>-SLAM: Inverting Imaging Process for Robust Photorealistic Dense SLAM*, in *ECCV* (2024).
7. **Changwoon Choi**, Jaeah Lee, Jaesik Park, and Young Min Kim, *3Doodle: Compact Abstraction of Objects with 3D Strokes*, in *SIGGRAPH (ACM TOG)* (2024).
8. Sang Won Im\*, Dongsu Zhang\*, Jeong Hyun Han, Ryeong Myeong Kim, **Changwoon Choi**, Young Min Kim\*\*, and Ki Tae Nam\*\*, *Investigating Chiral Morphogenesis of Gold Using Generative Cellular Automata*, in *Nature Materials* (2024).
9. **Changwoon Choi\***, Juhyeon Kim\*, and Young Min Kim, *IBL-NeRF: Image-Based Lighting Formulation of Neural Radiance Fields*, in *Pacific Graphics (Computer Graphics Forum)* (2023).
10. Sang Min Kim, **Changwoon Choi**, Hyeongjun Heo, and Young Min Kim, *Color Transform Module for Robust Novel View Synthesis*, in *Pacific Graphics (Computer Graphics Forum)* (2023).
11. Junho Kim, **Changwoon Choi**, Hojun Jang, and Young Min Kim, *LDL: Line Distance Functions for Panoramic Localization*, in *ICCV* (2023).
12. **Changwoon Choi**, Sang Min Kim, and Young Min Kim, *Balanced Spherical Grid for Egocentric View Synthesis*, in *CVPR* (2023).
13. Junho Kim, Hojun Jang, **Changwoon Choi**, and Young Min Kim, *CPO: Change Robust Panorama to Point Cloud Localization*, in *ECCV* (2022).
14. Dongsu Zhang, **Changwoon Choi**, Inbum Park, and Young Min Kim, *Probabilistic Implicit Scene Completion*, in *ICLR* (2022, *Spotlight*).
15. Junho Kim, **Changwoon Choi**, Hojun Jang, and Young Min Kim, *PICCOLO: Point Cloud-Centric Omnidirectional Localization*, in *ICCV* (2021).
16. Dongsu Zhang, **Changwoon Choi**, Jeonghwan Kim, and Young Min Kim, *Learning to Generate 3D Shapes with Generative Cellular Automata*, in *ICLR* (2021).

(\*Equally contributed.)

## Experiences

|                           |                                                                                             |                 |
|---------------------------|---------------------------------------------------------------------------------------------|-----------------|
| Research Scientist Intern | Meta Reality Labs, London                                                                   | 2025.06-2025.12 |
| Research Intern           | NAVER Cloud, Seoul                                                                          | 2023.12-2024.03 |
| Visiting Researcher       | UCSD, hosted by Hao Su                                                                      | 2023.05-2023.12 |
| Conference Reviewer       | SIGGRAPH, SIGGRAPH Asia, Pacific Graphics, NeurIPS, CVPR, ICCV, ECCV, 3DV, WACV, ACCV, BMVC |                 |
| Journal Reviewer          | ACM TOG, IEEE TVCG, Computer Graphics Forum                                                 |                 |

## Honors and Awards

|                                                            |             |
|------------------------------------------------------------|-------------|
| Oustanding reviewer award (CVPR 2026)                      | 2026        |
| Meshy Ph.D. fellowship finalist ( <a href="#">link</a> )   | 2026        |
| Outstanding poster presentation award (IPIU 2025)          | 2025        |
| Young researcher award (SNU INMC)                          | 2025        |
| Brain Korea 21 Ph.D. fellowship (NRF)                      | 2024 spring |
| Ph.D. Study-abroad program scholarship (SNU OIA, \$10,000) | 2023 spring |
| University outstanding T.A. award                          | 2022 spring |
| Samsung humantech paper award, Honorable mention (\$2,000) | 2021        |
| Merit-based scholarship (SNU, Full-Funding)                | 2015 - 2020 |

## Invited Talks

|                                                                                                        |                   |
|--------------------------------------------------------------------------------------------------------|-------------------|
| <i>"Abstraction and Style in 3D Shapes"</i> @ <a href="#">Disney Research Studios</a>                  | 2025.11.          |
| <i>"Computer Vision, Computer Graphics, and AI"</i> @ <a href="#">Seoul Science High School</a>        | 2025.04., 2026.04 |
| <i>"3Doodle"</i> @ <a href="#">Korea Computer Graphics Society (KCGS)</a>                              | 2024.07.          |
| <i>"Towards Robust Indoor Visual Localization with Omnidirectional Camera"</i> @ <a href="#">42dot</a> | 2022.05.          |

## Patents

- KR 10-2813248, KR 10-2969785

## Teaching

|                                                          |                               |
|----------------------------------------------------------|-------------------------------|
| Basic Calculus 1 (Tutor)                                 | 2015, 2016, 2018, 2020 spring |
| Basic Calculus 2 (Tutor)                                 | 2018 fall                     |
| Basic Physics 1 (Tutor)                                  | 2015, 2016, 2019 spring       |
| Basic Physics 2 (Tutor)                                  | 2015, 2019 fall               |
| Programming Methodology (Tutor)                          | 2019 fall                     |
| Engineering Mathematics 2 (Tutor)                        | 2019 fall, 2021 spring        |
| Signals and Systems (Tutor)                              | 2020 spring                   |
| Theory and Lab of IoT, AI, and Big Data (T.A.)           | 2019 fall, 2020 spring        |
| Signals and Systems (T.A.)                               | 2021 spring                   |
| 3D Computer Vision Track for AI Experts (Samsung) (T.A.) | 2021, 2022 summer             |
| Introduction to Random Variables and Processes (T.A.)    | 2022 spring                   |
| Physics Lab (T.A.)                                       | 2022 fall                     |